

AGROZEN — BROYLER FERMERI UCHUN AQLLI AI YORDAMCHI

Jo'ja kelgan kundan sotuvgacha — 42 kun davomida har kuni nima qilishni aytadi (Ross-308 / Cobb-500). Yangi: sensorlar, AI-kamera, bank hisoboti.

O'zbekcha · Русский · English | Android ilova · Veb-ilova | Internetsiz ham ishlaydi | Dori yozmaydi — veterinarga yo'naltiradi

NamDTU
Sun'iy intellekt yo'nalishi
STUDEXPO-2026

01 LOYIHA MAZMUNI

Boshlovchi broyler fermerlari uchun mobil yordamchi. Partiya kelgan sanani kiritsangiz, ilova har kuni **harorat, yem, suv va vazn** maqsadini ko'rsatadi, yozuvlarni tekshiradi va xavfli holatda ogohlantiradi. AI maslahatchi matn yoki ovozga uch tilda javob beradi.

QR-kodni skanerlang
ilovani yuklab oling







02 ISHLASH TARTIBI

1. **Til va hudud** — Uch tildan biri, tuman GPS orqali
2. **Partiya** — Kelgan sana va jo'jalar soni
3. **Bugungi reja** — 1–42 kun: harorat, yem, suv, vazifalar
4. **Kundalik** — Nobudlik, yem, vazn — bir bosishda
5. **Ogohlantirish** — Nobudlik >1% yoki vazn −15% bo'lsa
6. **AI + veterinar** — Savol bering yoki vetga qo'ng'iroq

AI SUN'IY INTELLEKT ASOSIDA ISHLAYDI

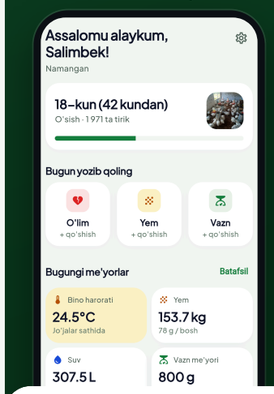
AI **faqat tekshirilgan bilimlar bazasidan** (10 mavzu, 3 tilda) javob beradi va partiyangiz kuni, soni hamda bugungi me'yorni hisobga oladi. Internetsiz — lokal bazadan. Dori va doza aytmaydi — veterinarga yo'naltiradi.

Jo'jalarim isitgich tagiga to'planib qolyapti, nima qilay?

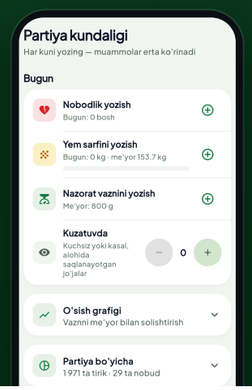
Bu sovuq belgisi. 3-kunda bino harorati **31,5°C** bo'lishi kerak — isitishni oshiring va jo'jalar teng tarqalganini kuzating.
Manba: AgroZen bilimlar bazasi · Ross-308

03 ILOVA KO'RINISHLARI — haqiqiy ekranlar: 18-kun, 2 000 boshli partiya

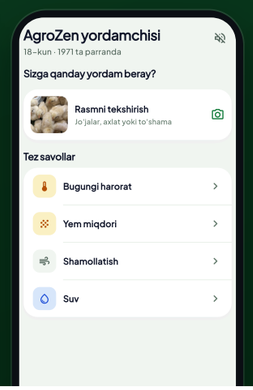
Bugun — Kun holati va me'yorlar



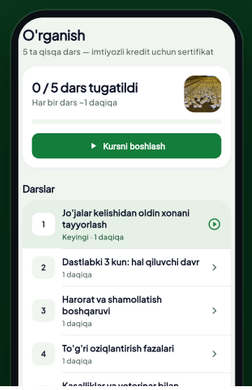
Kundalik — Bir bosishda yozuv



Yordamchi — AI chat va rasm tahlili



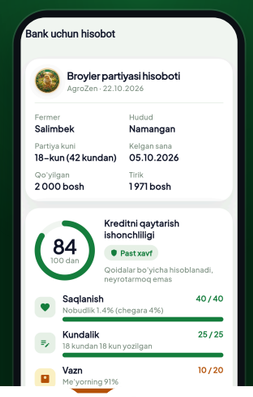
O'rganish — 5 dars va sertifikat



Aqlli ferma — Sensorlar, demo rejim



Bank hisoboti — Ball va PDF



04 ASOSIY IMKONIYATLAR

- **42 kunlik reja** — Ross-308 bo'yicha har kun
- **Partiya kundaligi** — Nobudlik · yem · vazn
- **Aniq ogohlantirish** — Qat'iy formula, taxmin emas
- **Ovozli AI maslahat** — Gapiring — tushunadi, o'qib beradi
- **Rasm tekshirish** — Jo'ja, axlat va to'shama holati
- **Ob-havo va mikroiqlim** — Tashqarida va bino ichida
- **Aqlli ferma** — Sensorlar, AI-kamera, avto-shamollatish
- **Tuman veterinari** — 13 hudud, bir bosishda qo'ng'iroq
- **Kurs va sertifikat** — Kredit olishdan oldin 5 dars
- **Bank uchun hisobot** — Fermer roziligi bilan, PDF

05 AQLLI FERMA — SENSORLAR VA AI-KAMERA — prototip: ilovada demo rejim, 2027-yilda ikki pilot fermada



- **Sensorlar** — ESP32: harorat, namlik, NH3, CO2, suv, tarozi
- **Edge-AI** — Jetson Orin Nano + IP-kameralar
- **Me'yorlar** — Ross-308 bo'yicha oddiy kod, AI emas
- **Harakat** — Ogohlantirish, veterinar, ventilyator

NH3 10 ppm gacha | CO2 3 000 ppm gacha | Ventilyator avtomatik yoqiladi | Harakatsiz parrandani kamera topadi

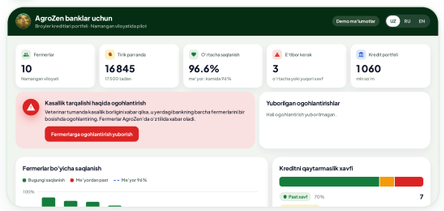
06 NATIJA — RAQAMLARDA

42	kun — har kun uchun tayyor reja	3	til — uz · ru · en, to'liq	5	dars — + tayyorgarlik sertifikati
13	hudud — veterinarlar ro'yxati	0	dori nomi — xavfsizlik qoidasi	260	test — avtomatik tekshiruv

07 TEXNOLOGIYA

- **Flutter 3 + Dart** — Android va veb ilova
- **Gemini 2.5 Flash** — AI javoblar (RAG)
- **Lokal RAG** — Internetsiz javoblar
- **Supabase** — PostgreSQL, Edge Functions
- **Speech-to-Text · TTS** — Ovozli savol va javob
- **Open-Meteo** — Ob-havo va prognoz

08 BANKLAR UCHUN — fermer roziligi bilan



- **Hisobot — PDF** — Kundalikdan tuziladi, faqat fermer rozi bo'lsa yuboriladi
- **Ball 0–100** — Saqlanish, kundalik, vazn, o'qish — qat'iy qoidalar
- **Kasallik haqida xabar** — Tumandagi barcha fermerlarga bir bosishda

09 NIMA UCHUN MUHIM

- **Yangi fermer yolg'iz emas** — Har kuni aniq ko'rsatma — tajriba bo'lmasa ham
- **Muammo erta seziladi** — Nobudlik va vazn har kuni me'yor bilan solishtiriladi
- **Bank ishonadi** — Kurs, sertifikat va partiya hisoboti

Ilova dori va doza tavsiya qilmaydi. Kasallik belgisida — tuman veterinariga qo'ng'iroq.

10 YO'L XARITASI

- **Hozir** — Android va veb-ilova, 3 tilda; aqlli ferma demo, bank hisoboti va paneli
- **Keyingi qadam** (reja) — Namangan viloyatida 10 fermer va bank bilan pilot; ikki fermada sensor va kameralar
- **Keyin** (reja) — Tuxum tovuqlari, qo'y va qoramol, issiqxonalar

Muallif: Zamalitdinov Salimbek (NamDTU, 4-kurs, 37-ATT guruhi) · Kodirov Ulug'bek

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github.com/ZedRubilan/agrozen

Har kuni aniq reja • AI maslahat — 3 tilda, ovozli • Xavfli holatda darhol ogohlantirish • Sensorlar va AI-kamera — kecha-kunduz • AgroZen — fermer cho'ntagidagi yordamchi